Fixed Geometric Health Decisions After LiDAR Degeneracy: A Controlled Transfer Study in Two LIO Backends

Harshil Patel, Daksha Lingampeta, and Nisarg Vyas

***Abstract*—Geometric health signals describe measurement constraints, but applications may interpret their decisions as evidence that local motion estimates have become accurate after degeneracy. We evaluate that interpretation for fixed policies transferred from development layouts to fresh simulated straight corridors in FAST-LIO and Point-LIO. The revised, geometry-screened design contains 48 matched corridor/control pairs in four length/width groups and 192 completed native runs. Each backend exposes a minimum-eigenvalue signal and a directional DCReg detector adaptation. Scalar thresholds are selected separately on development data and remain fixed during final evaluation. Reference-defined recovery requires three consecutive one-second motion windows within 0.20 m translation and 5° rotation error. FAST-LIO recovers in 46/48 corridor events; its scalar policy declares recovery in 14/46, with one initial false-healthy event among 46 eligible intervals. The Point-LIO directional adaptation produces 16/47 initial false-healthy events and declares 25/32 recoveries. Exploratory diagnosis separates misses caused by absent raw health, failed confirmation and an already-active state without a new declaration. These observations expose different fixed-policy trade-offs and distinguish active health from recovery announcements. The contribution is a reproducible transfer assessment of geometry-based decisions against sustained motion accuracy, conditional on the screened simulation design.**

***Index Terms*—LiDAR-inertial odometry, geometric degeneracy, recovery assessment, diagnostic reliability, threshold transfer.**

I. Introduction

LiDAR-inertial odometry (LIO) estimates motion from range and inertial measurements when global positioning is unavailable. Existing estimators such as FAST-LIO2 [1] and Point-LIO [2] provide motion estimates under varied operating conditions. In a long corridor, however, many measurements constrain lateral or vertical motion more strongly than translation along the corridor. A diagnostic describing these geometric constraints can inform an application, but its operational meaning requires careful evaluation.

Three quantities must be distinguished: measurement geometry, short-window motion accuracy, and accumulated pose drift. New visible structure may improve current constraints without erasing an earlier position error. Conversely, a conservative geometric rule can continue rejecting health while measured local motion already meets a specified accuracy criterion. Neither possibility alone is a new theoretical result. The empirical question is how often fixed decisions agree with the accuracy claim an application intends to make, after calibration is transferred to new layouts.

Prior work already investigates localizability, degeneracy mitigation and recovery. OCE evaluates diagnostic agreement with ground-truth motion usability and includes recovery behavior [3]; DCReg characterizes directional ill-conditioning and provides registration mitigation [4]. Consequently, recovery timing, false reassurance and geometric dependence cannot be claimed as newly introduced concerns. We examine a narrower setting: unchanged development-calibrated health policies evaluated on fresh, geometry-screened corridor dimensions in two separately instrumented LIO measurement stages.

The study uses a controlled simulator to isolate reference motion from the online estimators. Every selected corridor has a feature-added control, with identical motion and serialized IMU measurements. The primary endpoints measure premature confirmed health, new recovery declarations, detected-only delay and diagnostic availability. An exploratory investigation then examines the decision behavior behind misses without changing the original endpoints.

The paper makes three empirical contributions: (i) a matched two-backend evaluation of existing diagnostic families at fixed operating points; (ii) event and paired-error estimates with explicit geometry-level uncertainty and complete sample accounting; and (iii) a post-result diagnosis that distinguishes absent health from an already-active state without a new declaration. Together, these components show why score discrimination, fixed-point health and recovery announcements must be interpreted separately when a diagnostic is transferred to a new geometry range. The contribution is an empirical assessment and an auditable event-level explanation, rather than a new estimator or diagnostic formula.

II. Related work

a) Geometric localizability and mitigation.: X-ICP integrates directional localizability assessment with constrained registration [5]. SuperLoc predicts alignment risk for map-based LiDAR-inertial localization [6]. DCReg uses Schur complements to characterize rotational and translational ill-conditioning and couples this characterization with mitigation [4]. These systems establish that geometric information and its directional structure are important. Our experiment evaluates a DCReg detector adaptation on backend-specific measurement rows; it does not execute the complete DCReg system or reproduce its full characterization and mitigation pipeline.

b) Usability, transitions and recovery.: OCE derives an optimization-convergence diagnostic and evaluates motion usability against reference data, including premature recovery

and confirmation behavior [3]. This is the closest overlap with the present question. Informed, Constrained, Aligned examines degeneracy-aware registration in challenging environments and transitions [7]. ALIVE-LIO enhances degenerate-direction estimation using learned inertial velocity within an ESKF framework; its revised supplement also examines binary decision switching and short degenerate intervals [8]. Temporal stability itself is therefore also established prior work. The present study measures the transfer behavior of selected fixed policies; it does not establish that those problems were previously unrecognized.

c) Environmental variation and evaluation scope.: AdaLIO adapts parameters to corridor-like conditions [9], while GEODE provides diverse recorded degeneracy scenarios [10]. Our straight-corridor simulation supplies controlled motion and analytical reference rather than comparable real-world diversity. Table I summarizes the principal distinctions. None of these named systems was benchmarked against our policies under a shared evaluation, so the results cannot establish superiority over them.

III. Problem Formulation and Outcome Definitions

A. Local accuracy and accumulated drift

Let $T^r_{GB}(t)$ map comparison-body coordinates to the reference world, and $T^e_{WB}(t)$ map that same body to the estimator world. For associated times t_0, t_1, define

$$\Delta_r = T^r_{GB}(t_0)^{-1} T^r_{GB}(t_1), \tag{1}$$

$$\Delta_e = T^e_{WB}(t_0)^{-1} T^e_{WB}(t_1), \tag{2}$$

$$E_{\mathrm{rel}} = \Delta_r^{-1}\Delta_e = \begin{bmatrix} R_{\mathrm{rel}} & p_{\mathrm{rel}} \\ 0 & 1 \end{bmatrix}. \tag{3}$$

Translation error is $\|p_{\mathrm{rel}}\|_2$; rotation error is $\arccos(\mathrm{clip}((\mathrm{tr}\, R_{\mathrm{rel}} - 1)/2, -1, 1))$. Relative error does not require a fitted world-frame alignment. Accumulated drift is evaluated separately using a single pre-entry rigid alignment, with unit scale and no post-exit realignment. It does not define the primary health endpoint.

Primary windows use non-overlapping nominal integer-second boundaries. Each boundary is associated once to the nearest valid estimator pose within ± 0.05 s; ties select the earlier pose and adjacent windows share the same associated boundary. Actual durations must lie in $[0.9, 1.1]$ s. Reset-crossing windows, missing endpoints and unavailable reference are not replaced by zero error. Reference association and frame conversion are performed offline.

B. Recovery and its retrospective timing

The independently defined exit time t_e is the prescribed body crossing of the corridor exit plane. It is a geometric anchor, not an assertion that observability returns instantaneously. A one-second window passes if both translation error is at most 0.20 m and rotation error at most 5°. These operational tolerances adapt DCReg's registration-pair success reference [4] to LIO motion windows; they are not identical metrics or physical safety requirements.

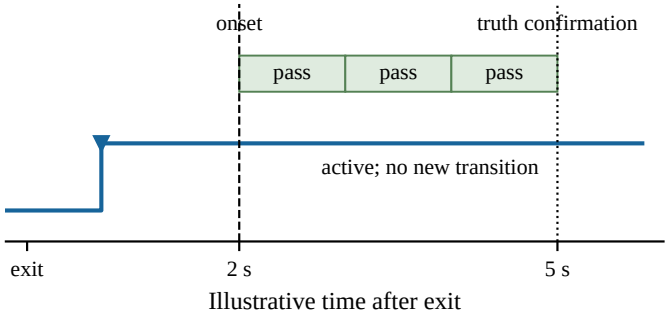


Fig. 1. Conceptual example, not a native result. Three passing motion windows establish onset retrospectively. Health active before onset can persist without a new declaration. The online policy receives no future reference windows.

Recovery is the first triple of consecutive passing post-exit windows whose actual endpoints all lie by $t_e + 20$ s. Onset t_o is the first window's actual start and truth confirmation t_c is the third window's end. Onset is therefore retrospective. A later failed or unavailable required estimated-motion window defines relapse at that window's start. The primary episode ends at the first relapse or the 20 s deadline. A completed event without a qualifying triple is a non-recovery; crashed or incomplete runs remain execution failures. This label does not establish that each event previously lost accuracy before exit.

C. Causal health and a new declaration

The health state machine runs on a 10 Hz grid before and after exit. At a decision tick it uses only the latest diagnostic record at or before that tick, at most 0.2 s old and in the same estimator segment. A newer invalid record cannot be bypassed to reuse an older healthy value. Three distinct healthy source records on consecutive ticks produce a transition into confirmed health. Unhealthy, unavailable or cross-segment evidence clears confirmation. Persistent health emits no repeated transition.

Initial false-healthy means confirmed health is active in the post-exit interval before t_o, or before the deadline if no recovery occurs. Its conditional denominator includes eligible intervals containing an available decision tick; the unconditional rate retains every non-empty initial interval. This endpoint concerns sustained recovery and need not imply a large instantaneous error at each premature tick.

A recovery declaration must be a new confirmed-health transition at or after t_o and before the recovery episode ends. Its delay is declaration time minus t_o. An already-active state may overlap recovered motion without generating a qualifying declaration. Such an event is a missed *declaration*, not necessarily persistent unhealthy output. Figure 1 illustrates the distinction.

D. Why active health and a declaration differ

Let $z_j \in \{0, 1\}$ be confirmed health at an available decision tick. The notification event is

$$a_j = z_j(1 - z_{j-1}). \tag{4}$$

TABLE I
SCOPE COMPARISON. THE EMPIRICAL QUESTION IS FIXED-POLICY TRANSFER AGAINST SUSTAINED LOCAL ACCURACY; THE NAMED SYSTEMS WERE NOT RUN AS SHARED-INPUT BASELINES.

Work	Established concern	Relation to this study
OCE	Ground-truth usability, false alarms and recovery behavior	Strong overlap; our question concerns fixed-policy transfer in two LIO backends.
DCReg	Directional detection, characterization and mitigation	We evaluate only an adapted detector on each backend's rows.
ICA	Degeneracy mitigation and challenging transitions	Our endpoints concern causal decisions against sustained local accuracy.
X-ICP / SuperLoc	Localizability and alignment risk	Relevant foundations; not interchangeable implemented baselines here.

If health remains active from before a recovered interval through its last tick, every a_j inside that interval is zero, although health is active throughout. Thus a state-based permission and a new-transition notification are different endpoints. This follows directly from the state machine; it is not evidence that every missed declaration remains unhealthy. The supplement provides the full argument and exact interval conventions.

IV. DIAGNOSTIC EXTRACTION AND FIXED POLICIES

A. Backend-specific measurement geometry

Read-only instrumentation records accepted point-to-plane measurement rows without feeding reference poses to either estimator. For rows ordered as translation then rotation, let $H = \sum_i j_i^\top j_i$ and accepted count N. The normalized scalar diagnostic is

$$G_L = \frac{S_L H S_L}{N\,0.001}, \tag{5}$$

$$S_L = \mathrm{diag}(1, 1, 1, L^{-1}, L^{-1}, L^{-1}), \tag{6}$$

$$s = \lambda_{\min}(G_{3\,\mathrm{m}}). \tag{7}$$

The 3 m lever scale and normalization are fixed protocol choices. This is measurement geometry, excluding filter priors, rather than calibrated posterior uncertainty. In particular, the denominator does not assert an identical native measurement-noise weight in Point-LIO.

FAST-LIO exports the last accepted scan linearization. Point-LIO changes state during point-time updates; its accepted groups are transported to a common world tangent basis before pooling. For captured pre-update rotation R_g, use

$$j_{g,\mathrm{common}} = [j_{t,g},\ j_{r,g} R_g^\top], \tag{8}$$

$$H_{\mathrm{frame}} = \sum_g j_{g,\mathrm{common}}^\top j_{g,\mathrm{common}}. \tag{9}$$

Point-LIO's pooled score is not a scan-end posterior information matrix. Each backend retains its own correspondences, poses, segments and calibration. Frame-end output is used for the common grid; this experiment does not test Point-LIO's maximum published output bandwidth.

B. Directional detector adaptation

For the adapted DCReg detector, reorder the raw six-dimensional Hessian into rotation and translation blocks and calculate

$$S_R = H_{RR} - H_{Rt} H_{tt}^\dagger H_{tR}, \tag{10}$$

$$S_t = H_{tt} - H_{tR} H_{RR}^\dagger H_{Rt}. \tag{11}$$

The pseudoinverse relative rank tolerance is 10^{-12}, an explicit adaptation setting. Subspaces are symmetrized and checked numerically. For each, directions with $\kappa_i = \lambda_{\max}/\lambda_i > 10$ are flagged; rank-deficient directions have explicit unbounded ratios. Raw health means no direction is flagged. This retains the fixed detector rule, without DCReg's full registration or mitigation [4].

At least six accepted rows and a finite numerically valid aggregate are required. A valid rank-deficient matrix remains geometry evidence, possibly with zero minimum eigenvalue. Startup, insufficient rows, invalid numerics and resets are explicit unavailable states. Missing or unfinished frames are not silently treated as healthy measurements.

C. What a geometry score cannot guarantee

A well-conditioned measurement matrix alone cannot bound estimation error without assumptions on residual error. In the illustrative linear model $y = Jx^\star + b$ with full-column-rank J, least squares gives

$$\widehat{x} - x^\star = (J^\top J)^{-1} J^\top b. \tag{12}$$

Choosing $b = Jv$ leaves the Hessian and its geometric scores unchanged but gives error v, which is unrestricted without an assumption on b. This standard linear counterexample motivates measuring motion accuracy separately. It does not identify the physical cause of errors in our native runs; the supplement states its assumptions and proof explicitly.

D. Development-only scalar calibration

Each backend selects one global scalar threshold using its own 32 formal corridor development events, seeds 14–45. Candidates exhaust finite unique valid scores plus an explicit no-alarm choice. Among candidates with at most 10% conditional initial false-healthy events, maximize declaration sensitivity and resolve ties toward the higher threshold. If no

feasible positive-sensitivity point exists, report that outcome explicitly. Controls and isolated feasibility replays do not fit thresholds.

The locked thresholds are 166.16366016039856 for FAST-LIO and 123.80876977779258 for Point-LIO. The directional rule remains fixed. Neither final scores nor final errors changed the thresholds. The development ceiling is a selection constraint, not a final population reliability bound.

V. Experimental design and statistical analysis

A. Controlled sensor input and matched scenes

The simulator uses a right-handed, z-up world and opaque rectangular surfaces. A room precedes and follows an axis-aligned corridor. The formal trajectory starts at $x = -6$ m, includes rest and a smooth acceleration ramp, then cruises at 0.8 m/s with prescribed smooth three-dimensional motion and attitude. Each scene lasts 60 s and provides the full 20 s post-exit horizon.

The generic LiDAR has 16 elevations between -15° and 15°, 360 azimuth columns and 10 Hz scans with point acquisition spread over 0.1 s. Rays use their acquisition-time poses and nearest-surface intersections. Range-noise standard deviation is 0.01 m. The IMU samples at 200 Hz; white-noise standard deviations are 0.002 rad/s and 0.02 m/s^2. Seeded initial gyro/accelerometer bias standard deviations are 0.003 rad/s and 0.03 m/s^2, with random-walk increments of 0.0001 rad/s/$\sqrt{\text{s}}$ and 0.001 m/s^2/$\sqrt{\text{s}}$. These are explicit stress settings, not calibration of a physical sensor. Extrinsics and clocks are ideal; reflectivity and weather are not modeled.

Figure 2 shows one recorded geometry. The control adds partial baffles along the positive corridor wall while preserving the open route. In the revised design the baffle count is $\max(4, \lceil L/(2\,\text{m})\rceil - 1)$, evenly spaced so longitudinal gaps are at most 2 m. Paired scenes share motion, reference and byte-identical IMU. Both backends receive matching corridor and control sensor-bag identities.

B. Development, geometry selection and exposure

Development layouts use lengths 9–13 m and corridor half-widths 1.6–2.4 m. The fresh evaluation uses four fixed groups (Table II); room half-width is sampled between 4.5 and 7.5 m. The geometry-only screen requires an axial-normal return fraction no larger than 0.10 in the corridor, at least 0.15 in the control, and at least 0.20 in the post-exit corridor scene five seconds after exit. These are eligibility proxies, not direct accuracy or observability labels.

An earlier geometry-only screen failed to supply eligible groups. A dated protocol amendment revised control spacing and reserved fresh candidate pools while keeping operating policies and primary outcomes fixed. Original IDs 100–179 were excluded. The first 12 eligible fresh IDs in ascending order within each group were selected: 48 pairs from 64 checked candidates. Screening preceded final estimator execution. The design therefore uses fresh instances but is not a wholly unadapted original test: the earlier geometry screen informed the revision. An earlier screen-digest discrepancy remains disclosed in the retained record and is not resolved by the fresh cohort. No final accuracy or diagnostic outcome selected a geometry.

TABLE II
Final geometry ranges. Width is twice the reported half-width. Each selected geometry has corridor/control scenes and two backend runs per scene.

Group	Length (m)	Half-width (m)	Pairs
Short/narrow	8–9	1.3–1.5	12
Short/wide	8–9	2.5–3.0	12
Long/narrow	15–18	1.3–1.5	12
Long/wide	15–18	2.5–3.0	12

C. Statistical units and uncertainty

The final schedule contains 48 geometry clusters, two scenes and two backends, yielding 192 native slots. Diagnostic ticks, two method rows and matched backend outputs are not extra independent geometries. Event proportions use 95% Wilson intervals. Group estimates retain 12 sampled geometries per group.

For the paired interior translation endpoint, planned one-second windows lie in the central half of the corridor. Each scene contributes its median error; each geometry contributes corridor median minus control median. The overall estimate is the equal-weight mean of the four group means. Uncertainty resamples paired geometry seeds within fixed groups, preserving pairs, for 10,000 PCG64 bootstrap draws with seed 20260926; percentile endpoints use NumPy's linear method. The analysis environment is Python 3.12.14 and NumPy 2.5.3. Intervals are conditional on these design strata.

Delay is summarized only among qualifying declarations, alongside the detection fraction. Misses receive no invented delay and non-recovery remains censored at the deadline. Unavailable denominators remain undefined, not zero. The protocol requires all planned usable pairs for the complete-data analysis; technical failures remain in the ledger and cannot be replaced by favorable scenes. No new significance test is introduced here.

VI. Results

A. Completion and reference-defined recovery

Both analyses completed all 48 geometry pairs and loaded 96 native scenes per backend, for 192 completed slots. All corridor events have usable reference. No native replay failed, and no native retry or repeat replaced a final slot. Two operational failures remain separately recorded: Point-LIO preflight stopped before any scene attempt because a scratch directory was absent, and the original primary analyzer rejected extra shared-threshold metadata before loading final reference or poses. A reviewed metadata-only adapter resolved the latter without changing the scientific analysis source.

FAST-LIO met the recovery rule in 46/48 corridor events (95.8%; Wilson 95% interval 86.0–98.8%); Point-LIO did so in 32/48 (66.7%; 52.5–78.3%). There were two and sixteen

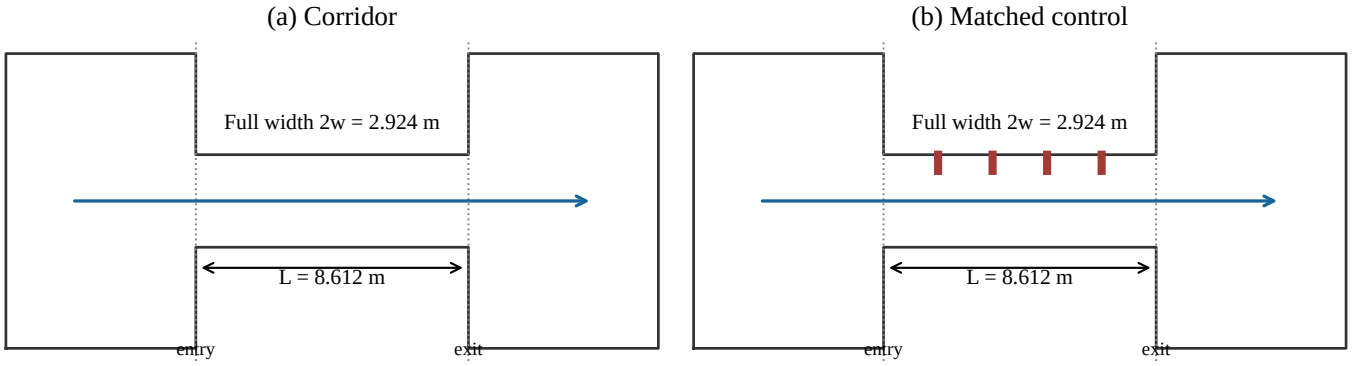


Fig. 2. Plan-view schematic of the recorded short/narrow layout, seed 10000. The control adds partial baffles without changing the route. Length and full width are the saved dimensions. Walls and control spacing follow the revised generator; the drawing omits room baffles, height, sensor rays and small three-dimensional motion for readability. It is not a reconstruction from estimator poses.

TABLE III
LOCKED POLICY OUTCOMES. FH IS CONDITIONAL INITIAL FALSE-HEALTHY; DECLARATIONS ARE AMONG OWN-BACKEND RECOVERED EVENTS. DELAY CONDITIONS ON DECLARATIONS ONLY. UNCONDITIONAL FH COUNTS ARE 1/48, 4/48, 4/48 AND 16/48.

Backend	Policy	Initial FH	Declarations	Misses	Delay (s)
FAST-LIO	Scalar	1/46	14/46	32	0.850
FAST-LIO	Fixed Schur	4/46	38/46	8	6.800
Point-LIO	Scalar	4/47	27/32	5	3.400
Point-LIO	Fixed Schur	16/47	25/32	7	3.000

non-recoveries respectively, and one relapse per backend. These own-backend outcomes on matching inputs are descriptive; they do not establish general estimator superiority.

B. Locked diagnostic trade-offs

Table III and Fig. 3 report the four fixed policies. FAST-LIO scalar health produced one conditional initial false-healthy event in 46 eligible intervals, but declared only 14 of 46 actual recoveries. Its directional adaptation declared 38/46 with four premature-health events. Point-LIO scalar declared 27/32 recoveries and produced four initial false-healthy events in 47 eligible intervals. Its directional adaptation declared 25/32 and produced 16/47 premature-health events.

All four policies have 9,600/9,600 available decision ticks in the primary post-exit horizon. This does not imply correct decisions or guarantee a tick in an extremely short initial unrecovered interval; hence the conditional FH denominators of 46 and 47 differ from 48. Scalar FH upper confidence bounds exceed the development ceiling of 10%; the final sample establishes no population guarantee below that ceiling. FAST-LIO scalar's 0.850 s median delay conditions on just 14 declarations, with 32 missed recoveries. It cannot be interpreted as superior overall responsiveness.

C. Geometry groups and paired interior error

Table V retains the full recovery denominators. Point-LIO recovered in 24/24 narrow versus 8/24 wide corridors. Its four scalar initial false-healthy events occurred in long/narrow layouts (4/12), while FAST-LIO scalar declared only 1/12 long/wide recoveries. Four-group summaries were predeclared; emphasizing these particular patterns afterward is exploratory, and no causal width effect is established.

The equal-group mean corridor-minus-control interior translation difference was +0.194 m [0.086, 0.311] for FAST-LIO and +0.149 m [0.068, 0.230] for Point-LIO (geometry-bootstrap 95% intervals). Positive aggregate effects do not establish a uniform benefit from control features: Point-LIO's short/narrow group mean was -0.087 m. The separate backend intervals are not a between-estimator superiority test. Relapse endpoints have only one event per backend; Point-LIO fixed Schur remains falsely healthy after that relapse (1/1), while the other policies have 0/1. These samples cannot support general relapse-reliability conclusions.

D. Score discrimination and label sensitivity

A high threshold-free score ranking does not guarantee a useful fixed-point recovery notification. FAST scalar has event-weighted ROC-AUC 0.921 [0.865, 0.955], yet declares only 14/46 recoveries at the locked operating point and causal confirmation rule. These measure different properties; we do not choose a replacement threshold from final data. The supplement reports ROC-AUC and average precision for every policy, with the already computed 10,000-draw geometry-bootstrap intervals.

Under the predeclared half, primary and double motion-error tolerances, FAST-LIO recovers in 44/48, 46/48 and 47/48 events, while Point-LIO recovers in 31/48, 32/48 and 33/48. These sensitivity results retain all corridors and do not replace the primary definition. The separate three-second rate-window analysis and complete stratum results are supplied in the supplement, with their different time and statistical units stated.

VII. EXPLORATORY FAILURE ANALYSIS AND DISCUSSION

A. Different observable reasons for a missed declaration

After the fixed analyses, an all-event investigation reused the frozen labels and state machine on all 96 corridor events. All

TABLE IV

WILSON EVENT INTERVALS. POLICIES WITHIN EACH BACKEND SHARE EVENTS; THESE ARE NOT INDEPENDENT-PROPORTION COMPARISONS.

Backend	Policy	Conditional FH (%; 95% CI)	Declarations (%; 95% CI)
FAST-LIO	Scalar	2.2 [0.4, 11.3]	30.4 [19.1, 44.8]
FAST-LIO	Fixed Schur	8.7 [3.4, 20.3]	82.6 [69.3, 90.9]
Point-LIO	Scalar	8.5 [3.4, 19.9]	84.4 [68.2, 93.1]
Point-LIO	Fixed Schur	34.0 [22.2, 48.3]	78.1 [61.2, 89.0]

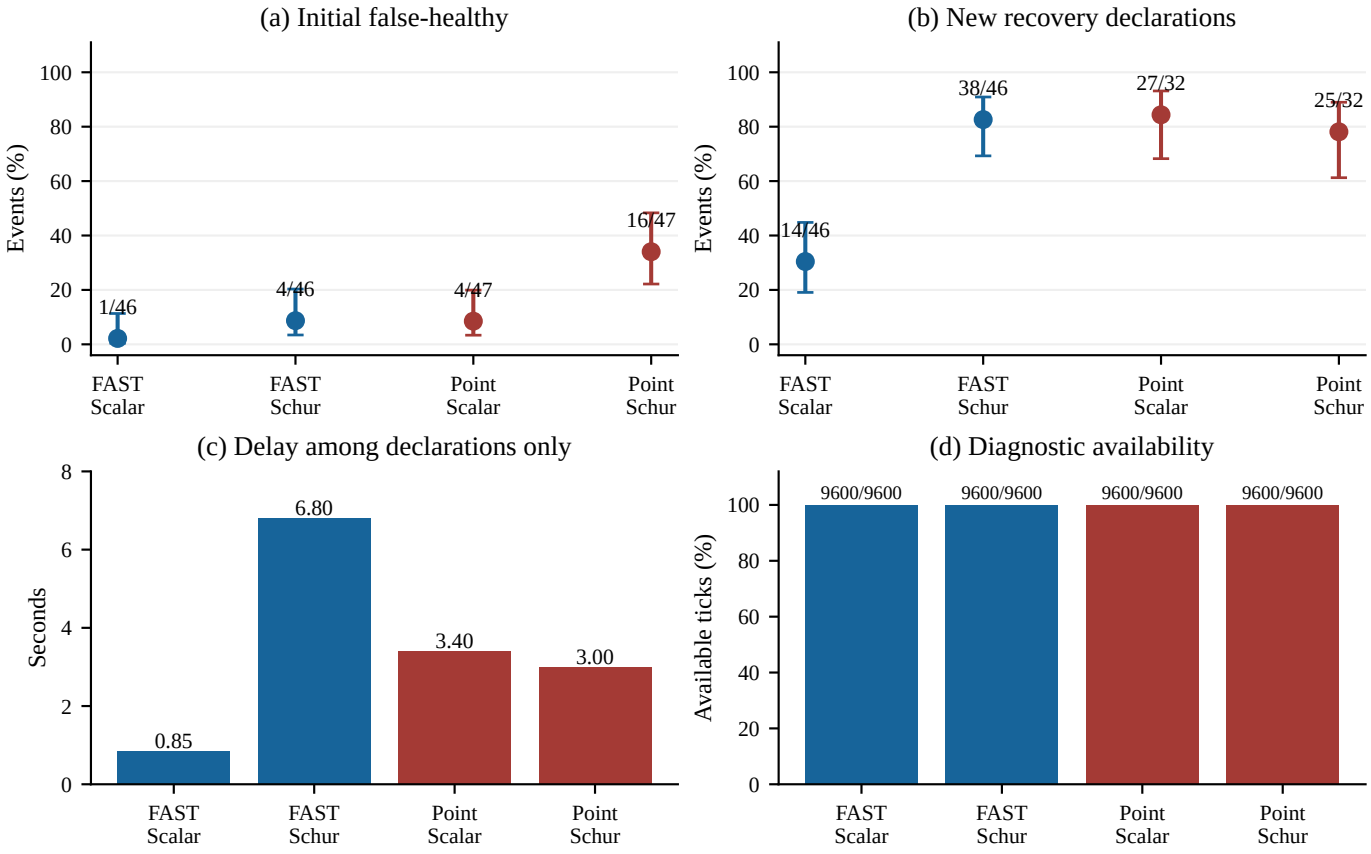


Fig. 3. The four locked policies: event rates with Wilson 95% intervals, detected-only delay, and diagnostic availability. FAST scalar declares only 14/46 recoveries despite its short conditional delay. Full score availability is not equivalent to reliable health decisions.

TABLE V

OWN-BACKEND RECOVERY UNDER THE FIXED RULE. EACH GROUP HAS 12 SHARED INPUT GEOMETRIES. ONE FAST-LIO LONG/WIDE AND ONE POINT-LIO SHORT/WIDE EVENT LATER RELAPSE.

	FAST-LIO		Point-LIO	
Group	Rec.	NR	Rec.	NR
Short/narrow	12/12	0	12/12	0
Short/wide	10/12	2	4/12	8
Long/narrow	12/12	0	12/12	0
Long/wide	12/12	0	4/12	8

TABLE VI

EXPLORATORY PARTITION OF EVERY MISSED DECLARATION. ACTIVE MEANS HEALTH WAS ACTIVE DURING SOME RECOVERED TIME WITHOUT A QUALIFYING NEW TRANSITION; CONTINUOUS HEALTH THROUGHOUT THE INTERVAL IS NOT REQUIRED.

Policy	No raw	No confirm.	Active	Total
FAST scalar	30	2	0	32
FAST Schur	5	2	1	8
Point scalar	3	0	2	5
Point Schur	2	0	5	7

192 diagnostic event rows were reproduced without changing cutoffs or outcomes. The mutually exclusive miss categories in Table VI describe observable decision behavior, not physical causes.

Figure 5 illustrates two of these distinct behaviors. Thirty FAST scalar misses have no raw healthy tick during the recovered episode; two have intermittent raw health without confirmation. In contrast, five of seven Point fixed-Schur misses involve active health without a new transition. Treating every miss as persistent rejection would erase this difference.

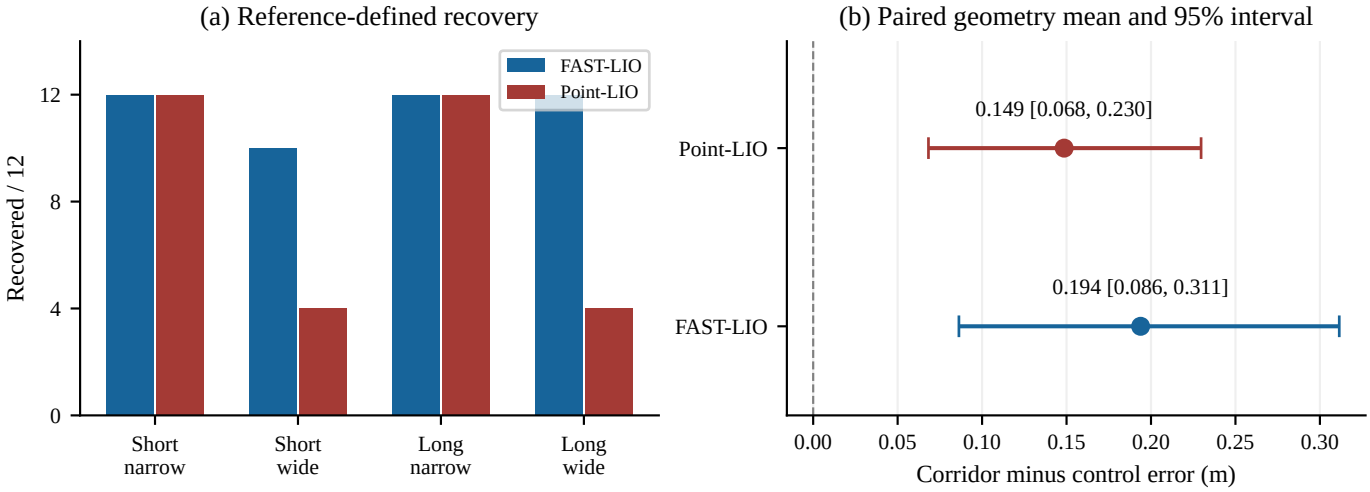


Fig. 4. (a) Recovery counts in all four fixed groups; 12 shared geometries per group. (b) Paired interior translation-error differences with the saved stratified geometry-bootstrap 95% intervals. The separate backend intervals are not a direct between-estimator comparison.

A continuous-permission application and an event-notification application require different interpretations of the same state trace. This observation motivates endpoint selection in future studies, rather than replacing the original endpoint with a more favorable result.

B. Measurement support does not supply a simple explanation

The investigation summarized accepted rows and scalar scores per event over the fixed post-exit horizon, then compared descriptive outcome groups. For FAST scalar, missed and detected recovered groups had almost identical group median accepted counts (397.50 and 397.25), but different group median scores (65.927 and 168.790). Point non-recovery events had a group median accepted count of 456.00, compared with 411.50 in recovered/detected events. These summaries do not support a simple monotonic count-only explanation. They do not rule out directional distribution, correspondence quality, visibility, IMU biases or map-history effects. Outcome grouping and the mechanistic questions were chosen after results and remain exploratory.

C. Interpretation relative to prior work

The findings complement existing usability and degeneracy work by documenting how selected fixed decisions behave under a specific transfer design. The observed trade-offs do not contradict a detector's geometric definition: measurement constraints are not identical to the history-dependent accuracy of an estimator. A short conditional delay, full score availability or a low point estimate of premature health is insufficient to characterize the complete policy. Counts, missed declarations and uncertainty must be reported together. OCE already treats usability and recovery [3]; our evidence should be judged for the additional insight from fixed-policy transfer and the observed failure distinctions, rather than the introduction of those evaluation concerns.

VIII. Limitations and Validity

The quantitative evidence is conditional on eligible rectangular straight corridors, four fixed dimension groups, a prescribed trajectory family and recorded sensor stress settings. Ideal timing/extrinsics and simplified opaque surfaces omit reflectivity, material and weather effects. Changing topology, motion, noise or initialization may change the results. The geometry screen conditions the sampled design; axial-normal coverage is not a general observability proof. The amended controls reflect information from a failed earlier screen, even though final estimator outcomes did not tune the policies.

There are 48 geometry clusters and only 12 per group. The emphasized width pattern is exploratory, and its cause has not been isolated. Each backend has its own instrumented measurement stage, settings and pose-derived labels; we do not infer a universal backend ranking. The directional comparator is an adaptation, not full DCReg, and no shared comparison with OCE, X-ICP, SuperLoc or other complete systems was run. The recovery label is a specified post-exit accuracy condition, not proof that accuracy was first lost in every event, and active-state usability differs from a new declaration.

Real-recording work does not close these limits. The Hilti-Oxford Exp18 development pilot had only 13/56 usable matched post-exit one-second windows and 0/56 three-second windows; its reference was partly LiDAR-map-derived. Earlier GEODE attempts were retained as failures. Neither supplies independent real-world sustained-recovery evidence for this paper. No physical aircraft was tested. Internal implementation and accounting reviews are scoped verification, not journal peer review or scientific novelty acceptance.

IX. Conclusion

Fixed geometric policies exhibited distinct premature-health and missed-declaration behavior when transferred to fresh, screened corridor layouts in FAST-LIO and Point-LIO. Low observed premature health could coexist with many missed

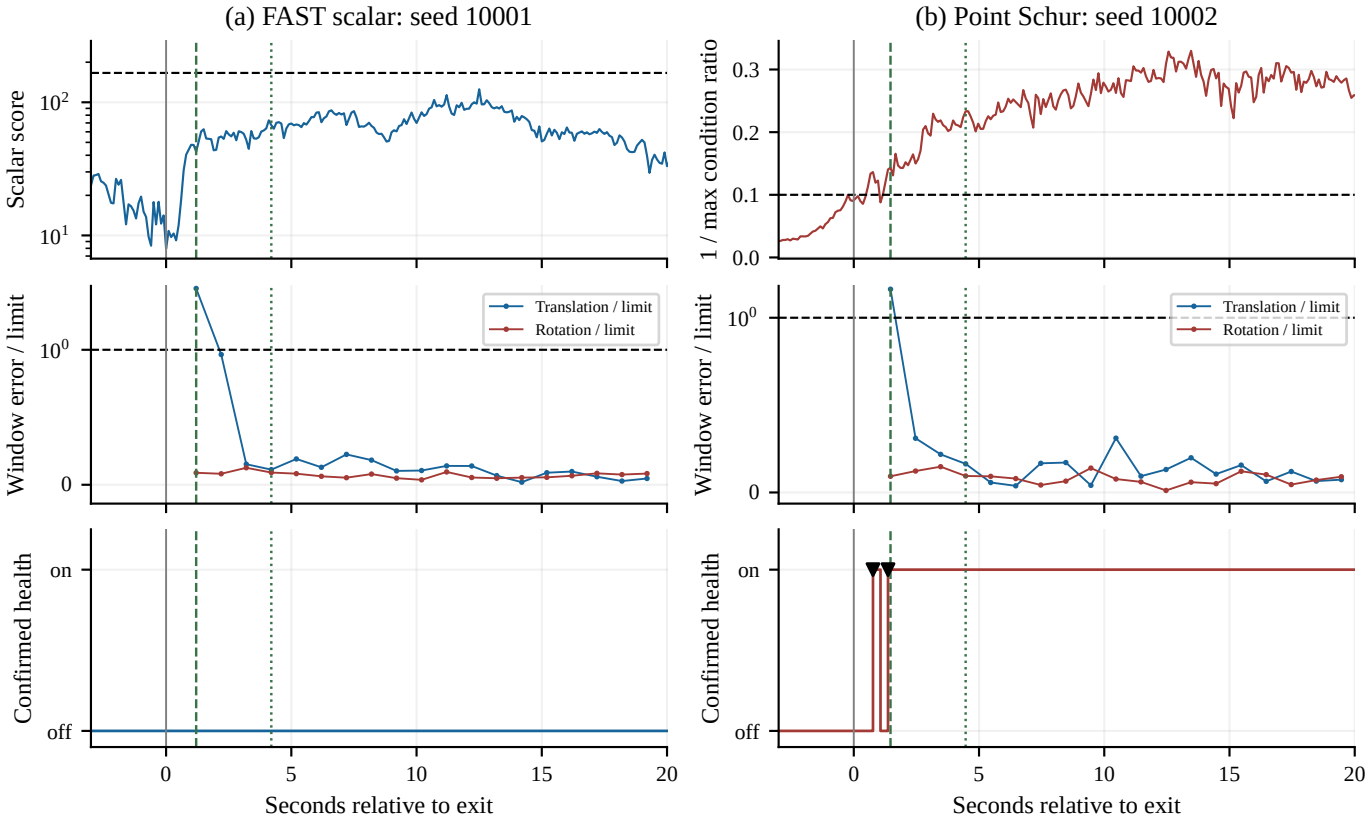


Fig. 5. Two native examples selected by the existing lowest-seed category rule, not by visual appeal. Left: FAST scalar miss without a raw healthy tick (seed 10001). Right: Point Schur miss with an already-active healthy state (seed 10002). Green dashed/dotted lines mark retrospective onset/confirmation; black triangles mark new online transitions. Window errors are shown at their actual end times and divided by the primary limits; they are not online inputs. Thin horizontal score lines are the fixed cutoffs. These examples illustrate the all-event partition, not its prevalence.

recoveries, while the fixed directional adaptation could produce substantial premature health in Point-LIO. Exploratory diagnosis showed that a missed declaration can reflect absent health, failed confirmation or an already-active state, with different operational implications. These controlled findings support evaluating diagnostic decisions against the specific accuracy and notification requirement of an application. Broader validation and a precise comparison with the closest literature are required before extending the conclusions beyond this design.

Reproducibility and Evidence Availability

The repository is https://github.com/Marshmellow31/gps-denied-drone-navigation. Compact evidence includes the final method/group CSVs, accepted analysis JSONs, event ledger, native receipts, figure manifests, protocol amendments, source/configuration identities and retained workflow failures under `research_paper/`. The evaluated geometry-v2 inventory binds 197 artifacts; a separately accepted metadata-compatibility supplement preserves the original scientific analysis. The manuscript is a later document and is not itself part of that scientific freeze.

The verified portable package is `reproduction/lidar_recovery_20261023/`, recorded at repository revision `d696dab`. It includes code, configurations, reviewed protocols, compact results, consumed native pose/health/reference streams and offline Python dependency installers. A relocated copy in a fresh offline-installed environment rebuilt four numerical reports and 20 figures, and reproduced all 96 recovery labels and 192 diagnostic event rows. Primary-window numerical error differences were at most 7.1×10^{-15}; labels and decisions matched exactly. The 197-entry implementation gate remained enforced. Current and exact frozen test versions are retained separately with restoration provenance.

The accompanying supplement adds exact native identities/configuration, all 64 revised screen candidates, sensitivity/ROC/AP reporting, and all 192 policy-event rows. Machine-readable native and event ledgers retain full precision. Raw bags, Jacobian sidecars and native binaries/environments remain external; replaying estimators from scratch is separate from stream reanalysis. No native experiment, threshold fitting, bootstrap rerun or new significance test was performed for this revision.

Author and Submission Information

The author names follow the existing project draft. Affiliations, corresponding author, contribution statement, funding

and competing-interest declarations require confirmation by the human authors. AI assistance was used in research tooling and manuscript preparation; the scope and required disclosure must be checked against the selected venue. All scientific claims and references remain the responsibility of the human authors. This working version uses the official IEEEtran conference class and its Times Roman text typography. A selected venue may require a different mode, page budget or anonymity treatment. It has not been submitted for peer review.

REFERENCES

[1] W. Xu, Y. Cai, D. He, J. Lin, and F. Zhang, "FAST-LIO2: Fast Direct LiDAR-inertial Odometry," 2021. https://arxiv.org/abs/2107.06829.

[2] D. He, W. Xu, N. Chen, F. Kong, C. Yuan, and F. Zhang, "Point-LIO: Robust High-Bandwidth Light Detection and Ranging Inertial Odometry," *Advanced Intelligent Systems*, vol. 5, no. 7, article 2200459, 2023. https://doi.org/10.1002/aisy.202200459.

[3] J. Zhang, M. Zhu, C. Sun, F. Guo, Q. Zhan, and R. Shen, "OCE: an optimization convergence metric for LiDAR SLAM degeneracy detection," *Autonomous Intelligent Systems*, vol. 6, article 21, 2026. https://doi.org/10.1007/s43684-026-00144-1.

[4] X. Hu, X. Chen, M. Jia, J. Wu, P. Tan, and S. L. Waslander, "DCReg: Decoupled Characterization for Efficient Degenerate LiDAR Registration," arXiv:2509.06285v3, 2026. https://arxiv.org/abs/2509.06285v3.

[5] T. Tuna, J. Nubert, Y. Nava, S. Khattak, and M. Hutter, "X-ICP: Localizability-Aware LiDAR Registration for Robust Localization in Extreme Environments," arXiv:2211.16335v4, 2024. https://arxiv.org/abs/2211.16335v4.

[6] S. Zhao et al., "SuperLoc: The Key to Robust LiDAR-Inertial Localization Lies in Predicting Alignment Risks," arXiv:2412.02901v2, 2025. https://arxiv.org/abs/2412.02901v2.

[7] T. Tuna, J. Nubert, P. Pfreundschuh, C. Cadena, S. Khattak, and M. Hutter, "Informed, Constrained, Aligned: A Field Analysis on Degeneracy-aware Point Cloud Registration in the Wild," arXiv:2408.11809v3, 2025. https://arxiv.org/abs/2408.11809v3.

[8] S. Kim, D. Lee, J. Hong, S. Park, H. Jo, and S. Han, "ALIVE-LIO: Degeneracy-Aware Learning of Inertial Velocity for Enhancing ESKF-Based LiDAR-Inertial Odometry," arXiv:2604.02706v2, 2026. https://arxiv.org/abs/2604.02706v2.

[9] H. Lim, D. Kim, B. Kim, and H. Myung, "AdaLIO: Robust Adaptive LiDAR-Inertial Odometry in Degenerate Indoor Environments," arXiv:2304.12577v1, 2023. https://arxiv.org/abs/2304.12577v1.

[10] Z. Chen et al., "Heterogeneous LiDAR Dataset for Benchmarking Robust Localization in Diverse Degenerate Scenarios," arXiv:2409.04961v2, 2024. https://arxiv.org/abs/2409.04961v2.